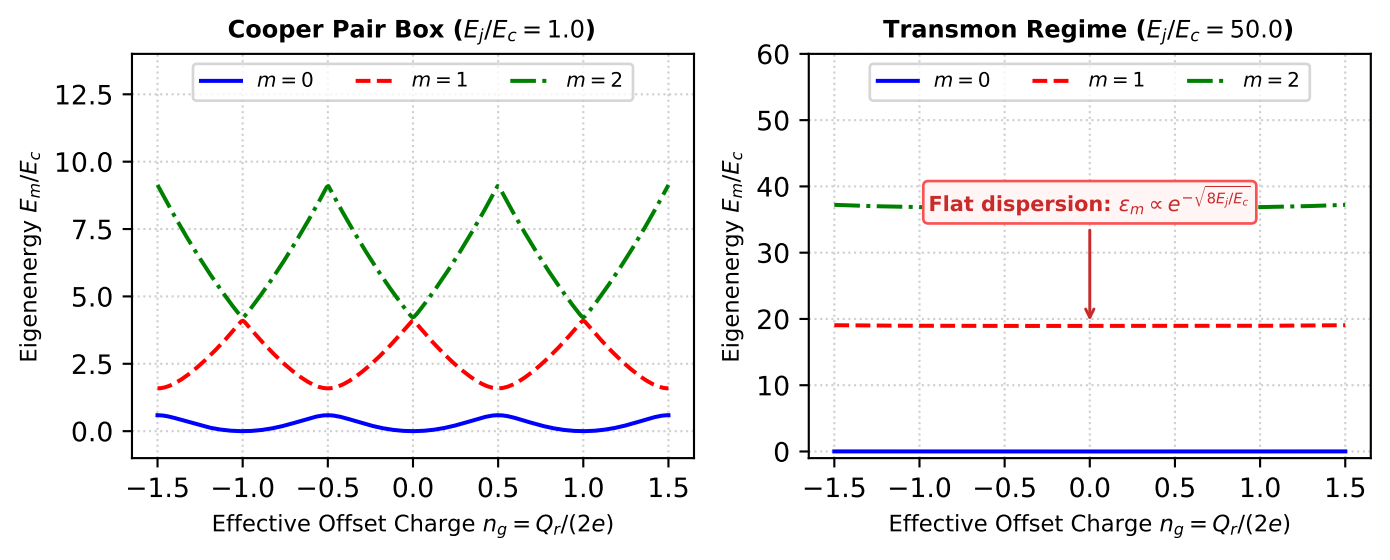
Cooper Pair Box ($E_j/E_c = 1.0$)
$m = 0$
$m = 1$
$m = 2$
Eigenenergy E_m/E_c
12.5
10.0
7.5
5.0
2.5
0.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Effective Offset Charge $n_g = Q_r/(2e)$
Transmon Regime ($E_j/E_c = 50.0$)
$m = 0$
$m = 1$
$m = 2$
Eigenenergy E_m/E_c
60
50
40
30
20
10
0
Flat dispersion: $\varepsilon_m \propto e^{-\sqrt{8E_j/E_c}}$
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Effective Offset Charge $n_g = Q_r/(2e)$